STATISTICS · SAMPLING AND EXPERIMENTATION

Sampling Methods Lab

Explore & Reflect

Name _____ Date _____ Period _____

Open **Sampling Methods Lab** in your unit's Interactive Math section on Mr. K's Classroom (or use your teacher's link). Answer Part 1 **before** you use the tool. Use the tool for Parts 2 and 3. Do Parts 4 and 5 on your own.

1 Predict before you touch the tool

1. A reporter asks the first 16 students who walk in the front door how much they slept. Will the average be accurate? Explain.

2. Does a larger sample fix a biased sampling method?

2 Explore use the tool

1. Take three simple random samples (n = 16) one at a time. Record each $\bar{x}$. Did any equal $\mu = 7.32$ exactly?

2. Convenience sample, n = 16: take 200+ samples. Record the center and the SD of $\bar{x}$. Explain both.

3. Convenience sample, n = 64: record the center. Did the bias go away?

4. With n = 16, compare 200+ SRS samples with 200+ stratified samples. Which is less variable? Why?

3 Record use the tool and fill in the table

Compare the methods ($\mu = 7.32$ hours)

Use 200 or more samples per row. Values vary a little; the key shows long-run values.

Method	n	Center of $\bar{x}$	SD of $\bar{x}$	Biased?
Simple random sample	16			
Stratified by grade	16			
Cluster (2 homerooms)	16			
Convenience	16			
Simple random sample	64			

4 Explain what did you discover?

1. A method is biased when its estimates are centered _____ the true value.
2. Larger samples reduce _____ but not _____.
3. Stratifying by a variable related to the response _____ variability.
4. Cluster sampling works best when each cluster _____.
5. Use the lab's results to explain the difference between bias and variability.

5 Apply a new situation, no tool

1. A school's four grades have very different sleep habits. To estimate average sleep with 80 students, which is better: an SRS of 80 or 20 random students from each grade? Why?

2. Identify the method: choose 5 of the school's 40 homerooms at random and survey everyone in them.

3. A news website poll gets 12,000 responses. Is the result trustworthy? Explain.